

GAT-CTDE: Federated Multi-Agent Reinforcement Learning with Graph Attention for Multi-gNB Slice Admission Control

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Abstract—Our prior work validated single-gNB, single-agent Deep Reinforcement Learning (DRL) slice admission control (SAC) on a live Open RAN testbed. This paper extends that control problem to a multi-gNB cluster, where per-gNB admission decisions must be coordinated rather than made independently. We propose GAT-CTDE, a centralised-training, decentralised-execution multi-agent architecture in which a shared Graph Attention Network (GAT) encoder over the gNB topology graph feeds per-slice Q-heads, and evaluate it against an independent per-gNB DQN ablation and a flattened-state single-agent DQN baseline over a 30-seed offline stress-regime campaign. We report a methodological finding alongside the architectural one: our initial GAT encoder, lacking any normalisation, collapsed to a trivial “accept every request” policy on all 30 seeds, and the standard episode-level SLA-compliance metric could not detect this, since it happened to score the collapsed policy no worse than several genuinely-differentiated ones. Two encoder fixes – per-layer LayerNorm, then per-slice Q-heads that isolate each slice’s gradient signal – reduce the collapse rate from 30/30 to 9/30 seeds, and we introduce a correctness-aware evaluation pair (mean reward per step and block precision, both derived from data every arm already logs) that the compliance metric cannot substitute for. On mean reward per step, GAT-CTDE decisively beats an independent-per-gNB-DQN ablation that isolates the same GAT/centralised-training contribution (+1.461, 95% CI [0.550, 2.782], Wilcoxon $p < 0.0001$, 27 wins / 0 ties / 3 losses), but its advantage over our existing single-agent baseline is small and not statistically decisive at this sample size (+0.195, 95% CI [−0.017, 0.442], Wilcoxon $p = 0.0577$, 20 wins / 3 ties / 7 losses out of 30 seeds) – a more modest finding than we originally reported, corrected after catching a second data-integrity bug with the same discipline the collapse diagnosis required (Section IV). We then build a federated variant, in which each gNB trains locally and synchronises via periodic FedAvg with a DP-SGD-style per-client gradient clip and calibrated Gaussian noise, and find that federating the architecture costs nothing measurable in reward at this sample size (+0.133, $p = 0.8125$), while differential privacy shows a real threshold effect on block precision – perfect targeting through noise multiplier $\sigma = 1.0$, dropping sharply by $\sigma = 4.0$ – rather than the compliance metric’s misleading, non-monotonic curve at the same noise levels. Finally, we evaluate the same frozen policies under mid-episode gNB dropout, demand spikes, and agent churn (one gNB’s decisions replaced by a fresh, never-trained policy): dropout and churn both produce a real, accelerating cost with severity (not graceful degradation) across every arm, while demand spikes do not threshold at all on block precision, a genuinely different failure mode. We cross-

check every result in this paper against an independent seed sample, not only the same seeds reproduced again, and report one finding this caught: an initial reading that independent-per-gNB DQN is architecturally immune to agent churn did not survive replication and is retracted here rather than reported. These results argue that admission-control evaluation in multi-agent, privacy-constrained, disruption-prone settings needs metrics tied to the actual training objective, not only to a downstream SLA proxy that can reward the wrong policy, and that architectural claims need more than one seed sample before they are trusted.

Index Terms—5G, Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Differential Privacy, Open RAN

I. INTRODUCTION

Network slicing lets a single physical radio access network serve heterogeneous service classes – Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC) – under distinct per-slice Service Level Agreements (SLAs). Our earlier work [1], [2] established that a Deep Q-Network (DQN) admission controller can learn to enforce these SLAs under a single gNodeB (gNB), and validated this on a live OpenAirInterface (OAI) testbed under a Quality-of-Experience-aware reward.

A real multi-cell deployment does not admit requests at one gNB in isolation: neighbouring cells share backhaul, spectrum planning, and – in the reward formulation we build on here – a cluster-wide load-balance objective. Coordinating admission decisions across gNBs is therefore a genuinely multi-agent problem, and the natural architectural question is whether a shared, graph-structured representation of the cluster’s state helps a set of per-gNB agents make better joint decisions than either a single agent using the flattened joint state, or independent per-gNB agents with no shared representation at all.

This paper makes four contributions. First, we build GAT-CTDE: a Graph Attention Network [3] encoder over the gNB topology, shared across agents and trained centrally, feeding decentralised per-slice Q-heads that each agent executes using only its own node’s embedding. Second, we report a methodological failure mode and its fix: the initial encoder, lacking any output normalisation, encoded network congestion almost entirely as embedding *magnitude* rather than *pattern*, causing training to collapse to a trivial always-accept policy – and the

standard SLA-compliance metric used throughout our prior work could not distinguish this collapse from genuinely correct behaviour, because blocking a low-priority slice under congestion, the reward-optimal action, costs compliance without ever recovering it once a stress episode is underway. We introduce two fixes and a pair of correctness-aware metrics that expose this gap directly. Third, we extend GAT-CTDE to a federated training regimen with differential privacy, and characterise both the cost of federating (statistically indistinguishable from zero at our sample size) and the cost of privacy (a real threshold in decision precision, not a smooth compliance trade-off), using the same correctness-aware metrics. Fourth, we evaluate every arm’s frozen policy under mid-episode disruption – gNB dropout, demand spikes, and agent churn – and cross-check every result in the paper against an independent seed sample rather than only reproducing the same seeds, which is what catches a real architectural overclaim (Section VIII) before it could be reported as a finding. We label these three campaigns M2 (centralised), M3 (federated/privacy), and M4 (disruption) throughout, matching this project’s internal milestone numbering; a fourth, earlier milestone (M1, live-testbed recalibration) motivates the offline stress environment of Section II but reports no results of its own here.

TODO(MANOJ): Literature review section deferred – to be written and inserted after this Introduction once the related-work survey is ready. The rest of the draft does not depend on it.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Cluster Topology and Admission Decisions

We consider a cluster of $N=3$ gNBs, each serving the same three slices (URLLC, eMBB, mMTC) under a shared per-gNB Physical Resource Block (PRB) budget B . As in our single-gNB work, an admission decision is realised as a PRB-ceiling adjustment (`slicing_control_m` min/max ratio), not a literal per-flow accept/reject at the scheduler, since the real OAI xApp control API only exposes ratio ceilings. No real multi-gNB physical topology (inter-site distances, backhaul graph, neighbour relations) is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected – every gNB attends to every other gNB’s state – and note this explicitly as a modelling choice rather than a measured fact.

At each control step, every gNB has zero or more pending admission requests, each tagged with a slice identity. An agent decides accept/reject for each pending request independently, conditioned on the current cluster state; the discrete action space matches our single-gNB work exactly (`action_dim=2`), so any coordination benefit found here is attributable to the multi-agent architecture, not to a richer action space.

B. Reward and the Differentiated-Shedding Regime

The reward combines a per-slice SLA-compliance term with a cluster-wide load-balance penalty, weighted by a per-slice `priority_weight` (URLLC highest, mMTC lowest) and a congestion coefficient that was tuned, in prior work, so that

the reward-optimal policy under congestion is *differentiated shedding*: continue accepting URLLC, mostly accept eMBB, and reject mMTC. This is not an assumption we make for this paper – it is a property of the existing reward weights, confirmed directly by Q-value inspection on a working baseline policy (Section IV), and it is the behaviour against which we judge whether a new architecture is doing what it is supposed to do, not merely whether it scores well on a downstream proxy.

C. Offline Stress-Regime Environment

All results in this paper are obtained in an offline, closed-loop simulation environment deliberately oversubscribed to roughly 110% of one gNB’s PRB budget across its three slices, reusing our existing multi-gNB configuration unmodified. A separate investigation (held out of this paper’s scope) found that this offline environment’s demand process does not reproduce the live testbed’s SLA-margin distribution closely enough to support a live-rank prediction claim, even after recalibrating the demand model’s temporal structure. We therefore use it here explicitly as a *stress environment* for the contention regime our live rig does not reach, not as a claim that any ranking found offline would reproduce live – the single live gNB validated in our prior work remains the only real-hardware anchor, and multi-gNB claims in this paper are offline-only throughout.

III. METHODOLOGY

A. GAT-CTDE: Shared Encoder, Decentralised Q-Heads

Fig. 1(a) shows the architecture. At every control step, each gNB’s own per-slice [PRB-used-ratio, congestion-level, queue-length-norm] features form that node’s row in a joint node-feature matrix, mirroring exactly the per-(gNB, slice) triple our existing multi-gNB state encoding already computes – kept as a per-node matrix here instead of flattened, since the GAT encoder needs graph structure, not a flat vector. A two-layer, four-head GAT [3], implemented directly (the small graph sizes here make a dense, masked-attention implementation as fast as a sparse one and avoid an external dependency), produces one embedding per gNB. Centralisation comes from this encoder: its parameters are updated by every agent’s loss jointly, so it sees the full cluster’s state and structure during training. Decentralised execution comes from what happens next: at action-selection time, each agent consumes only its own node’s embedding row to decide its own pending requests – never another agent’s row.

1) *Per-Slice Q-Heads*: A first version of this architecture fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head MLP. Under this design, mMTC’s true reward-optimal margin under congestion is small by the reward’s own calibration (a net expected value on the order of -1 against Q-values in the hundreds, a consequence of this architecture’s per-request temporal-difference scheme), while URLLC’s much larger priority weight produces proportionally larger gradients from URLLC-context samples. Because all three slices’ samples update the *same* shared parameters, URLLC’s abundant, large-magnitude gradients can overwrite

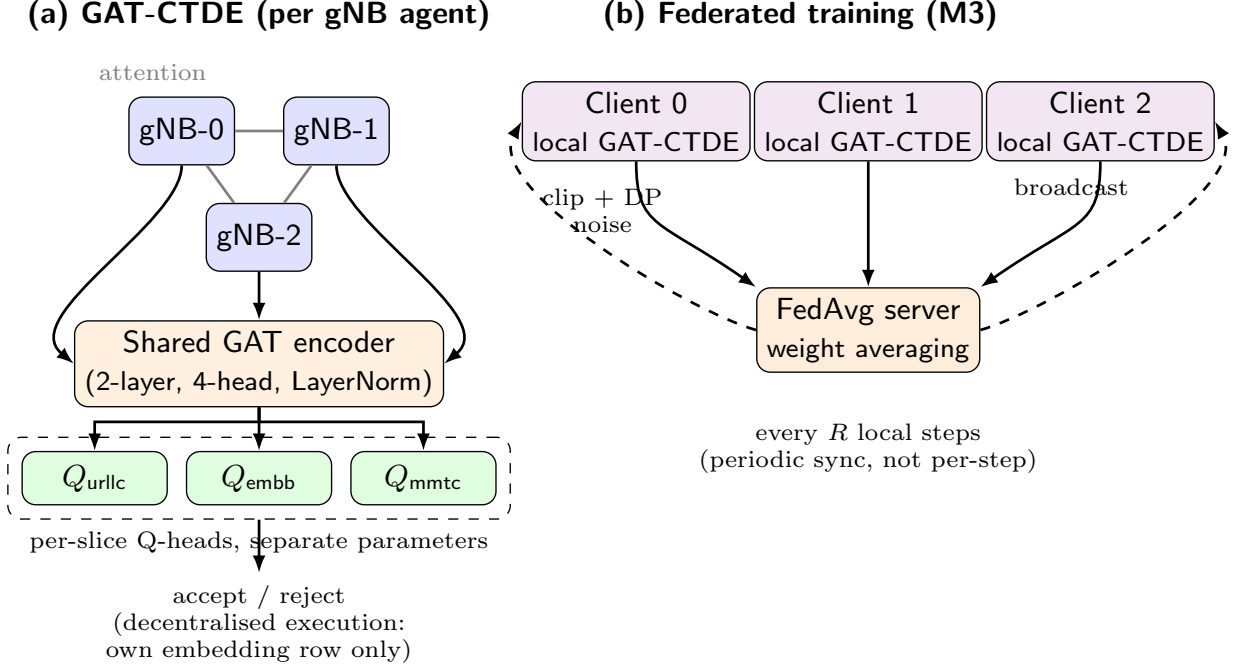


Fig. 1. GAT-CTDE architecture. (a) A shared Graph Attention encoder over the fully-connected 3-gNB topology produces one embedding per node; per-slice Q-heads, each with its own parameters, consume that node’s own embedding row at action-selection time (decentralised execution). (b) The federated training variant: each gNB is a client with a private copy of the same network, synchronised via periodic FedAvg with an optional per-client DP-SGD clip-and-noise step.

whatever fine adjustment mmTC’s much smaller true signal needs before it registers – a standard shared-parameter/class-imbalance failure mode. We replace the single shared head with *one small MLP per slice*, selected by the context one-hot at the point where the head is applied (Fig. 1(a)): each slice’s decision boundary now has its own parameters, so one slice’s gradients can no longer directly overwrite another’s. The shared GAT encoder upstream is unaffected and remains the architecture’s centralising element.

2) *Training Procedure*: Algorithm 1 states the training loop precisely: the encoder is updated from every request in a sampled minibatch, regardless of which gNB or slice it belongs to (centralised training), while action selection at rollout time consults only the acting agent’s own embedding row (decentralised execution). The per-request TD formulation on line 13 is the fix described above in the context of the architecture’s earlier, unstable QMIX-style variant (Section IV’s companion bug, documented in full in `ctde_policy.py`): every pending request in a step is its own independent temporal-difference sample, never summed or averaged across a step before the loss is computed, matching our existing single-agent DQN’s own granularity exactly.

B. Federated Training with Differential Privacy

Fig. 1(b) shows the federated variant. Each gNB becomes an FL client with a private copy of the same GAT-CTDE network (minus the unused QMIX-style mixer our centralised training does not require), initialised identically by a server broadcast. Every client trains only on its own requests – transitions are partitioned by owning agent, so no client ever sees another

client’s raw admission data – while its GAT encoder still consumes the full joint node-feature snapshot, the same public per-slice occupancy information every other arm in this paper observes. Every R local training steps, the server averages all clients’ online weights (FedAvg [4]; weighted equally, since every client runs the same amount of local training) and broadcasts the result back. We also implement FedProx [5], which differs only in the client-side objective (a proximal term pulling local weights toward the last broadcast global model), but do not run a full campaign with it: our topology treats every gNB as an interchangeable peer with no client heterogeneity for FedProx’s correction to address, so a full sweep would be evaluating a fix for a problem this environment does not have.

Differential privacy is applied as a standard DP-SGD [6] step: every client’s gradient is clipped to a fixed norm and, when enabled, perturbed with calibrated Gaussian noise scaled by a noise multiplier σ , immediately before that client’s own optimiser step – so the local update itself is privatised, not only the round-boundary aggregate. We report σ as the primary privacy parameter throughout. A formal (ϵ, δ) bound is also reported, computed via zero-concentrated differential privacy (zCDP) [7] composition of the Gaussian mechanism: with $\rho = k/(2\sigma^2)$ over k per-client releases, converted via $\epsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}$. This is a conservative, non-subsampled composition bound, not a tight moments-accountant figure – we report it alongside σ , not in place of it.

Algorithm 2 states one federated round precisely. Every client runs the identical per-request TD loss Algorithm 1 already defines (line 8’s FedProx term is additive and zero whenever $\mu=0$, our reported configuration); clipping is applied

Algorithm 1 GAT-CTDE: centralised training, decentralised execution

Require: shared GAT encoder f_θ , per-slice Q-heads $\{Q_{\phi_k}\}_{k=1}^K$, target networks $f_{\theta^-}, \{Q_{\phi_k^-}\}$, replay buffer $\mathcal{D}$, fully-connected adjacency A , discount γ , target-sync period C episodes

- 1: **for** episode $= 1, 2, \dots$ **do**
- 2: reset environment; observe joint node features $X_1 \in \mathbb{R}^{N \times d}$
- 3: **for** step $t = 1, \dots, T$ **do**
- 4: $E_t \leftarrow f_\theta(X_t, A)$ {one embedding row per gNB; centralised: sees every node}
- 5: **for** each pending request r at gNB i , slice k **do**
- 6: $a_r \leftarrow \arg \max_a Q_{\phi_k}(E_t[i])[a]$ w.p. $1-\epsilon$, else uniform random {decentralised: own row $E_t[i]$ only}
- 7: **end for**
- 8: $X_{t+1}, \rho_t, \text{done} \leftarrow \text{ENV.STEP}(\{a_r\})$
- 9: store $(X_t, X_{t+1}, \{(i, k, a_r)\}_r, \rho_t, \text{done})$ in $\mathcal{D}$
- 10: **if** $|\mathcal{D}| \geq \text{warmup size}$ **then**
- 11: sample minibatch $B \subset \mathcal{D}$
- 12: $E_B \leftarrow f_\theta(X_B, A)$; $E'_B \leftarrow f_{\theta^-}(X'_B, A)$ {target net, no grad}
- 13: **for** each request $(i, k, a) \in B$ **do**
- 14: $q \leftarrow Q_{\phi_k}(E_B[i])[a]$
- 15: $y \leftarrow \rho + \gamma \cdot (1 - \text{done}) \cdot \max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
- 16: **end for**
- 17: $\mathcal{L} \leftarrow$ mean Huber loss over every request's (q, y) pair {per-request, never summed per step}
- 18: update $\theta, \{\phi_k\}$ by $\nabla \mathcal{L}$; clip grad-norm to 1.0
- 19: **end if**
- 20: **end for**
- 21: **if** episode mod $C = 0$ **then**
- 22: $\theta^- \leftarrow \theta$; $\phi_k^- \leftarrow \phi_k \forall k$
- 23: **end if**
- 24: **end for**

identically whether or not noise is added (line 11 vs. line 9), so a $\sigma=0$ run and a $\sigma>0$ run differ only in whether noise is injected, not in any other aspect of the client's update – the isolation the privacy- cost comparison in Section VII depends on.

IV. A TRAINING COLLAPSE, DIAGNOSED AND FIXED

A. Discovery

While generating a privacy-utility sweep for the federated variant (Section VII), noise levels that should have produced genuinely different, stochastically-trained checkpoints instead produced bit-identical per-seed evaluation compliance. Tracing this back, we found that our first GAT-CTDE encoder's eval policy blocked zero requests across every evaluation episode, on all 30 campaign seeds – an always-accept policy is a functional no-op, so any two checkpoints that reach it produce identical environment trajectories regardless of their internal weights.

Algorithm 2 Federated GAT-CTDE round with DP-SGD

Require: N clients, each with local network $(\theta_i, \{\phi_{i,k}\})$; global snapshot $\bar{\theta}$; local steps per round R ; clip norm C_{dp} ; noise multiplier σ ; FedProx weight μ ($= 0$ unless noted)

- 1: **for** round $= 1, 2, \dots$ **do**
- 2: **for** local step $= 1, \dots, R$, each client i independently **do**
- 3: sample minibatch B_i of i 's own requests from i 's local buffer
- 4: $E \leftarrow f_{\theta_i}(X_{B_i}, A)$ {full joint state; client's OWN local weights only}
- 5: $\mathcal{L}_i \leftarrow$ per-request Huber TD loss (Alg. 1, lines 11–13) $+ \frac{\mu}{2} \|\theta_i - \bar{\theta}\|^2$ {FedProx term}
- 6: $g_i \leftarrow \nabla_{\theta_i, \phi_i} \mathcal{L}_i$
- 7: **if** $\sigma > 0$ **then**
- 8: $g_i \leftarrow \text{clip}(g_i, C_{\text{dp}})$; $g_i \leftarrow g_i + \mathcal{N}(0, (\sigma C_{\text{dp}})^2 I)$ {DP-SGD [6]}
- 9: **else**
- 10: $g_i \leftarrow \text{clip}(g_i, C_{\text{dp}})$ {same clip norm, no noise – isolates the privacy cost}
- 11: **end if**
- 12: $(\theta_i, \phi_i) \leftarrow (\theta_i, \phi_i) - \eta g_i$
- 13: **end for**
- 14: $\bar{\theta}, \bar{\phi} \leftarrow \frac{1}{N} \sum_{i=1}^N (\theta_i, \phi_i)$ {FedAvg [4], equal client weights}
- 15: **for** each client $i = 1, \dots, N$ **do**
- 16: $(\theta_i, \phi_i) \leftarrow (\bar{\theta}, \bar{\phi})$ {broadcast}
- 17: **end for**
- 18: **end for**

B. This Was Not the Reward-Optimal Policy

We confirmed, rather than assumed, that always-accept was a genuine failure and not a legitimate optimum: a direct Q-value probe on the collapsed checkpoint, under a synthetic congested state, showed $Q(\text{accept}) - Q(\text{reject})$ positive and large for every slice, including mmTC – the same broken signature our reward-tuning process had already documented and fixed once before, for a different architecture, under different (already-superseded) reward weights. Under the current, correctly-calibrated weights, the independent-DQN ablation (Section VI), which sees identical raw features through no encoder at all, does not exhibit this collapse – isolating the encoder, not the reward or the training loop, as the fault.

C. Root Cause: Magnitude, Not Pattern

Comparing the encoder's own embedding for a synthetic idle state against a congested one on the collapsed checkpoint: the raw input difference has ℓ_2 norm 2.55, and the resulting embedding difference has norm 19.6 – an apparently large, congestion-sensitive response. But the cosine similarity between the two embeddings is 0.99997: the encoder was encoding congestion almost entirely as embedding *magnitude*, with no mechanism anywhere in the GAT layers preventing this, and no rotation of direction to speak of. Since the downstream Q-head is a roughly linear function of its input,

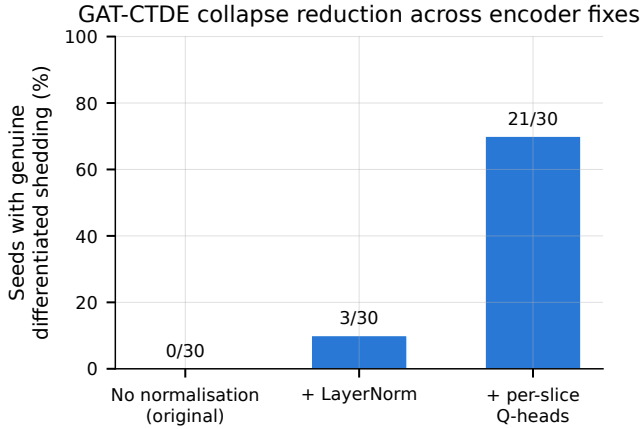


Fig. 2. Fraction of the 30 campaign seeds reaching genuine, correctly-targeted differentiated shedding (mmTC-only blocking under congestion), across the original unnormalised encoder and the two fixes described in Section IV.

larger magnitude pushes $Q(\text{accept})$ up faster than $Q(\text{reject})$ as congestion increases – backwards from the reward’s intent.

D. Fixes

We add a per-layer `LayerNorm` to the GAT encoder, constraining every node’s embedding to zero-mean, unit-variance regardless of input magnitude. This is a real, verified improvement – collapse drops from 30/30 to 27/30 seeds – but not a complete fix. We then apply the per-slice Q-heads described in Section III-A1, which reduces collapse further, to 9/30 seeds (Fig. 2). We verified each fix on a small pilot before committing to a full campaign’s compute budget, and we tested one further variant, disabling `LayerNorm`’s learnable affine transform, that made results *worse* (0/3 pilot seeds differentiated, versus 1/3 with the affine transform enabled) and was reverted – not every plausible fix helped, and we report the one that did not.

E. A Second Bug: Eval-Log Contamination

A separate, unrelated bug was caught while building an evaluation-time disruption harness for planned future work (Section X): our logging utility opens its output file in append mode and never truncates it, and GAT-CTDE was retrained in place three times over the course of diagnosing and fixing the collapse above (the original encoder, the `LayerNorm`-only fix, and the per-slice-heads fix), each time writing evaluation results to the same per-seed path. Because nothing cleared that path between runs, every one of the 30 seeds’ evaluation logs had silently accumulated all three runs’ episodes stacked together – confirmed directly by each log’s own episode index resetting to 1 twice. Every GAT-CTDE number reported through an earlier draft of this paper, and every M3 result that reuses GAT-CTDE’s centralised evaluation as a reference point (Section VII), was therefore a hidden three-way average across two architecturally-obsolete runs and the true final one, not the final architecture’s behaviour alone. The saved model checkpoints themselves were unaffected (a different save routine overwrites rather than appends, verified by reproducing

one seed’s final-block statistics exactly from its checkpoint in isolation), so the fix required no retraining: we cleared every stale log and re-ran evaluation once, cleanly, against the already-trained checkpoints, and hardened the training script to clear its output path before any future run. All numbers in this paper reflect that corrected data; the magnitude of the correction is largest on exactly the paired comparisons this section’s own diagnostic discipline exists to protect, which is why we report it here rather than silently revising the numbers.

F. Correctness-Aware Metrics

The fixes above make the architecture more correct while making its `sla_compliance_all_slices` score *worse* (Section VI): blocking mmTC under congestion, the reward-optimal action, does not recover mmTC’s own SLA margin in this stress regime, so every newly-correct decision costs compliance the metric never credits back. We therefore report two further metrics, both computed from data every arm already logs, requiring no new instrumentation:

- **Mean reward per step:** the actual training objective, matching our existing evaluation harness’s own established definition (per-episode mean of every step’s logged reward, then mean across episodes) – since the reward’s own calibration is what defines correct behaviour in the first place, this tests whether a policy does what it was trained to maximise, directly.
- **Block precision:** of every request an arm blocks, the fraction that target mmTC specifically, the only slice the reward calibration ever makes reject-optimal. Undefined, and reported as such, for seeds that never block anything.

V. EXPERIMENTAL SETUP

Both the centralised (Section VI) and federated (Section VII) campaigns use 300 training and 50 evaluation episodes per seed, matching our existing single-agent training budget so that no comparison in this paper trades training time for a favourable result. The centralised campaign uses a fixed 30-seed list (seeds 900–929) with three arms: GAT-CTDE; an independent-DQN ablation (one per-gNB DQN instance, no parameter sharing, no topology information, the identical per-agent input GAT-CTDE’s Q-head consumes) that isolates the GAT/centralised-training contribution specifically; and a single-agent DQN baseline, our existing architecture run unmodified against the flattened joint state, our today’s established multi-gNB pattern. All three arms use matched hyperparameters (discount factor $\gamma = 0.95$, per-episode epsilon decay), a parity fix applied after an earlier pass found the two new architectures had inadvertently inherited a raw-DQN default schedule instead of our tuned one. The federated campaign uses the first 10 of these 30 seeds, the same per-seed training budget, and a five-level noise-multiplier sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$, so its $\sigma=0$ level is directly comparable to the centralised campaign’s same 10 seeds without re-running that arm. All confidence intervals are 95% bootstrap percentile intervals (10,000 resamples, not a normal approximation, since compliance is bounded on $[0, 1]$ and empirically right-skewed);

TABLE I
M2 CAMPAIGN RESULTS, 30 SEEDS/ARM, 95% BOOTSTRAP CIs.

Arm	Compliance	Mean reward/step	Block precision
GAT-CTDE	0.151 [0.083, 0.225]	14.062 [13.91, 14.23]	0.952
Indep. DQN	0.049 [0.013, 0.093]	12.601 [11.30, 13.49]	0.901
Single-agent	0.115 [0.061, 0.176]	13.867 [13.69, 14.05]	0.987

paired comparisons use the Wilcoxon signed-rank test, matching this project’s established preference for nonparametric methods.

VI. RESULTS: CENTRALISED GAT-CTDE CAMPAIGN

Table I and Fig. 3 report the final, post-fix 30-seed campaign. On `sla_compliance_all_slices`, GAT-CTDE’s mean is 0.151, below its own pre-fix number (0.353) for the reason given in Section IV-F – the metric penalises newly-correct mmTC-blocking decisions – and the paired comparison against single-agent DQN shows no clear direction: +0.036 (95% CI [−0.050, 0.126], Wilcoxon $p = 0.4549$, 11 wins / 9 ties / 10 losses). On mean reward per step, the metric that reflects the actual objective, GAT-CTDE beats the independent-DQN ablation decisively – +1.461 (95% CI [0.550, 2.782], $p < 0.0001$, 27/0/3), isolating a real contribution from the shared GAT encoder and centralised training – but its edge over single-agent DQN is small and does not clear conventional significance at this sample size: +0.195 (95% CI [−0.017, 0.442], $p = 0.0577$, 20/3/7). Its block precision (0.952) is high regardless, meaning that when GAT-CTDE does block a request, it is almost always the reward-correct slice. 21 of the 30 seeds now show genuine, correctly-targeted differentiated shedding (20 of those 21 blocking mmTC exclusively), up from 3/30 under the LayerNorm-only fix and 0/30 originally – a real, substantial behavioural improvement that, at this sample size, does not by itself translate into a decisive reward edge over the single-agent baseline specifically.

VII. RESULTS: FEDERATED TRAINING AND DIFFERENTIAL PRIVACY

A. Federation Cost

Comparing the centralised GAT-CTDE campaign against the federated variant with no DP noise, on the same 10 seeds: mean reward per step moves from 14.073 to 13.940, a paired difference of +0.133 (95% CI [−0.138, 0.470], Wilcoxon $p = 0.8125$) – no measurable cost to federating at this sample size (5 of the 10 seeds tie exactly, and the remaining seeds split 2 centralised / 3 federated). Both arms’ block precision is perfect (1.000) on the seeds that block anything at all.

B. Privacy Cost: A Threshold, Not a Smooth Curve

Fig. 4(b) reports block precision across the noise sweep. Precision is perfect through $\sigma = 1.0$, then drops sharply – 0.834 at $\sigma = 2.0$, 0.584 at $\sigma = 4.0$: differential privacy does not degrade decision quality gradually here, it holds until a real threshold and then degrades quickly. Mean reward per step, by contrast, stays in a comparatively narrow band

across the whole sweep (12.45–14.19, with $\sigma = 2.0$ ’s own estimate also carrying the widest confidence interval of the five levels, [10.40, 13.99]), meaning DP noise does not crater the policy outright even at the highest tested σ – it erodes the *precision* of which slice gets blocked, not whether the policy tries to differentiate at all. The compliance-based sweep over the same data (not shown) is non-monotonic and its sign flips between adjacent σ levels; we do not report it as the primary privacy-utility curve for that reason, and note this as a second instance, independent of Section IV’s architectural finding, of `sla_compliance_all_slices` failing to track the quantity we actually care about.

VIII. RESULTS: DISRUPTION RESILIENCE

A. Setup

We evaluate every arm’s already-trained, frozen checkpoint (no retraining) against three mid-episode disruptions, injected purely at evaluation time: *gNB dropout* (one randomly-chosen gNB’s own requests are forced to reject and its row in the joint feature matrix every agent’s encoder reads is zeroed, for a window); *demand spike* (a transient multiplier on the arrival rate, for a fixed 30%-of-episode window); and *agent churn* (one gNB’s decisions are made by a freshly-initialised, never-trained copy of its own policy class instead of the loaded checkpoint, for a window). We call the third one *agent churn* rather than *client churn* deliberately: federation only happens during training, so a frozen, post-aggregation federated checkpoint has no per-client staleness left to reintroduce, and no arm shares cross-agent information at *inference* time beyond what each node’s own encoder already reads from the joint state – there is no FL-specific desynchronisation left to disrupt. Replacing one agent’s trained decision-making with an inexperienced stand-in is a different, well-motivated failure mode (active-but-uncoordinated vs. absent) that applies to every genuinely multi-agent arm; single-agent DQN, with no separable per-gNB agent, is excluded from this condition. Severity is the disruption window’s duration (10%/30%/60% of the episode) for dropout and churn, and the arrival-rate multiplier ($2\times/4\times/8\times$) at a fixed window for spike. Target gNB and window start step are drawn fresh from each episode’s own seeded random-number generator (RNG), not fixed to one gNB.

Because a demand spike changes request *volume*, raw mean reward per step is not a fair comparison against an undisrupted baseline – the reward’s own service term is linear in how many requests were accepted that step (Section III), so more requests mechanically inflates it regardless of decision quality. We therefore normalise by each step’s own already-logged pending-request count before averaging (mean reward per pending request), built from data every run already logs, and use this normalised metric as the primary comparison throughout this section. Fig. 6 shows exactly where each disruption kind intercepts the per-step control loop around the frozen policy.

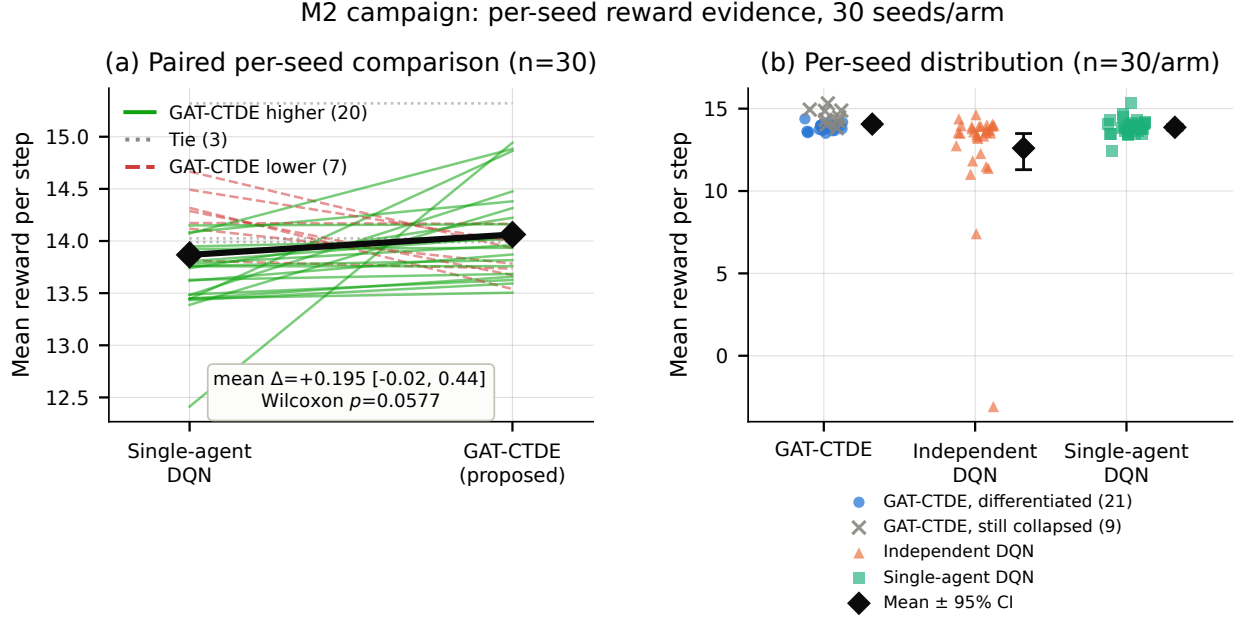


Fig. 3. 30-seed campaign, per-seed evidence for mean reward per step, the correctness-aware metric (Section IV-F). (a) Paired per-seed comparison against single-agent DQN: each line is one seed, coloured by whether GAT-CTDE scored higher (green, 20), tied (grey, 3), or lower (red, 7) than the baseline on that seed – the same paired unit the Wilcoxon test is computed over, and the source of this figure’s borderline $p = 0.0577$. (b) Full per-seed distribution for all three arms; GAT-CTDE is split into seeds that learned genuine differentiated shedding (blue, 21) versus seeds still stuck at the always-accept collapse (grey $\times$, 9) – shown without claiming the split predicts reward, since several still-collapsed seeds score among the arm’s highest (always accepting never pays the reject cost).

B. Dropout and Agent Churn: A Real, Accelerating Threshold Effect

Fig. 5(a)-(b) report the normalised-reward cost at each severity. Dropout costs every arm significantly at every severity (Wilcoxon $p \leq 0.03$, 9 or 10 of 10 seeds hurt in every case), and the cost accelerates with severity rather than growing proportionally: for GAT-CTDE, +0.165 at the 10% window, +0.601 at 30% (roughly $3.6\times$ the first increment), +1.710 at 60% (roughly $1.8\times$ the second increment) – the same super-linear pattern holds for every arm. Agent churn shows the same accelerating shape for GAT-CTDE (+0.059/+0.200/+0.529, $p = 0.0039$ throughout) and the federated arm (+0.073/+0.240/+0.585, $p = 0.0039$ throughout).

Independent DQN’s own churn cost needs both samples reported together to state honestly. On the seeds used throughout the rest of this paper (900–909), it is small and does not clear conventional significance (+0.000/+0.168/+0.640, $p = 0.084/0.106/0.065$) – which an earlier pass of this analysis read as independent DQN being architecturally immune to churn, reasoning that a policy which never attends to another agent’s features (confirmed by inspection: it reads only its own node’s row) has no coordination benefit a churned peer can take away. An independent-seed replication (seeds 1000–1009, disjoint from every other result in this paper) contradicts that reading directly: independent DQN’s churn cost there is significant at every severity (+0.097/+0.657/+1.223, $p = 0.0020$ throughout, 10/10 seeds hurt each time) and *larger* than GAT-CTDE’s or the federated arm’s at matched severities in that sample – the opposite of immune. The direction was

consistent across both samples throughout (churn always hurts, never helps); what was wrong was treating a borderline, underpowered $n = 10$ result as a confirmed architectural property. We do not conclude independent DQN is immune to churn: replacing a trained agent’s own decision-making with an untrained one plausibly degrades that agent’s local performance regardless of whether it ever used cross-agent coordination, and the combined evidence across both samples now points that way.

C. Demand Spike: A Different, Cleaner Story

Fig. 5(c) shows no threshold at all on block precision, which stays flat and near-ceiling through the highest tested multiplier for every arm. The normalised-reward signal (not shown) that spike *improves* per-request reward is real and reproduces across both seed samples for GAT-CTDE, independent DQN, and the federated arm ($p \leq 0.014$ in both), but we do not read it as better decision-making: it traces to a specific, understood property of the existing reward shape (a fixed per-step violation penalty diluted by a larger denominator once request volume rises, still present after the primary volume normalisation above, which only corrects the dominant, much larger effect from the reward’s accept-scaling service term) – block precision, a volume-invariant ratio, is the metric that actually answers whether spike corrupts decision quality, and it says no. Single-agent DQN’s version of this same reward-based signal is significant in the original sample ($p = 0.0020$ at every severity) but not in the replication ($p = 0.084$) – direction consistent, significance sample-dependent, reported as such rather than as an unconditional finding.

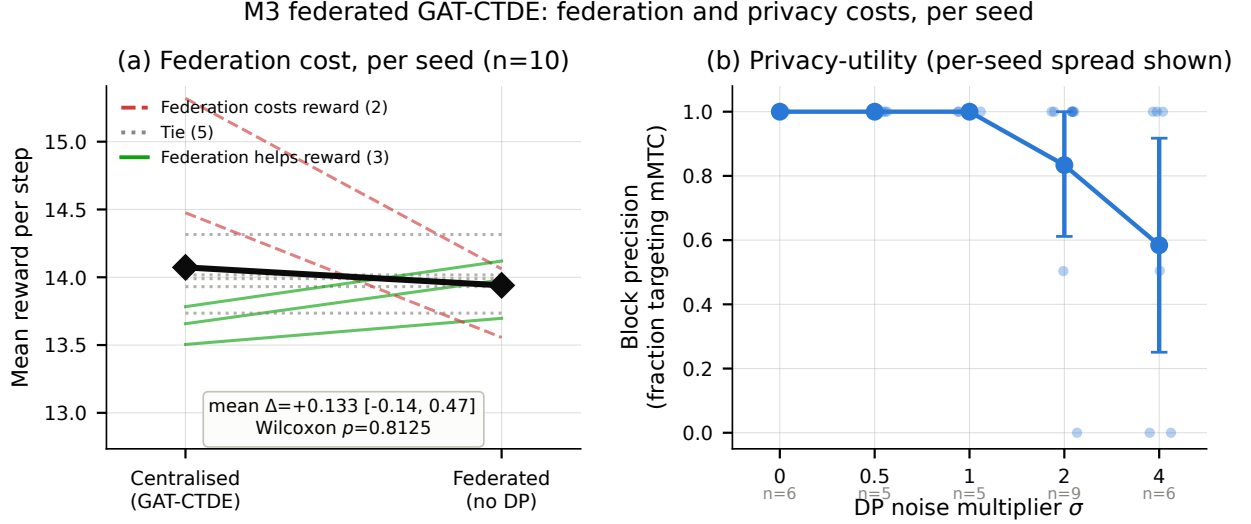


Fig. 4. Federated GAT-CTDE, per-seed evidence. (a) Paired per-seed federation cost on mean reward per step (same 10 seeds): centralised training scores higher on 2 seeds, federated on 3, with 5 ties – no measurable difference (Wilcoxon $p = 0.8125$), shown as individual seed trajectories rather than two independent bars, since the significance test itself is paired. (b) Privacy-utility curve using block precision, the metric that survives the compliance-based curve’s non-monotonic, uninterpretable pattern at the same noise levels: individual seed values (light dots) are shown behind the mean, along with the seed count that ever blocked anything at each σ (precision is undefined otherwise) – precision is perfect through $\sigma = 1.0$ and drops sharply by $\sigma = 4.0$.

M4: disruption cost vs. severity, 10 seeds/arm

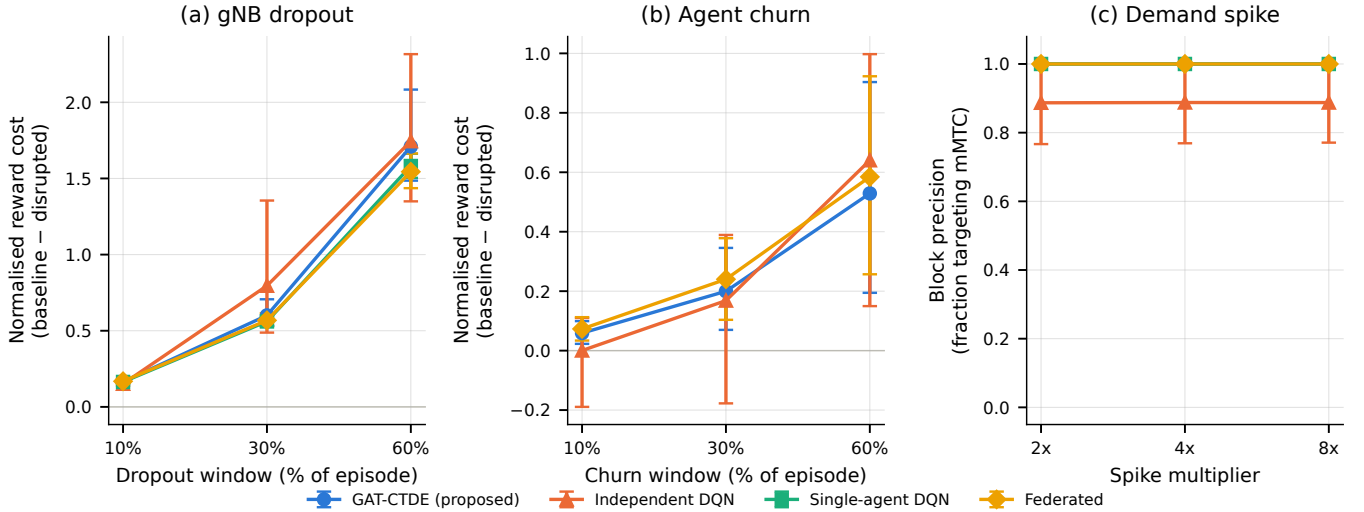


Fig. 5. M4 disruption campaign, 10 seeds/arm. (a)-(b) Normalised reward cost (baseline – disrupted) vs. severity for dropout and agent churn – every arm, every severity, shows a real cost that accelerates with severity rather than growing linearly. (c) Block precision vs. demand-spike multiplier: flat and near-ceiling for every arm, unlike (a)-(b) – spike does not threshold on the metric that matters.

D. Independent-Seed Replication

Every dropout result above, and GAT-CTDE/the federated arm’s churn costs, were checked against the independent 1000–1009 sample and hold up in both direction and approximate magnitude – e.g. GAT-CTDE’s dropout costs there are $+0.170/+0.552/+1.449$, close to the 900–909 sample’s $+0.165/+0.601/+1.710$. The one finding that did not survive (independent DQN’s supposed churn immunity) is reported only in its corrected form above, not as an initial claim we later walk back – this section already reflects what replication

showed, exactly the discipline Section IV-E argues for. One caveat the replication surfaced on the M3 privacy sweep (Section VII) rather than here: block precision’s degradation *exists* in both samples but the exact noise level it begins at does not (perfect through $\sigma = 1.0$ in one 10-seed sample, already degrading at $\sigma = 0.5$ in the other) – a threshold is real; its precise location is not a property we would claim generalises beyond the specific ten seeds it was measured on.

Fig. 7 puts every headline paired comparison in this paper next to its independent-sample counterpart on one axis, recomputed directly from each sample’s own raw evaluation logs

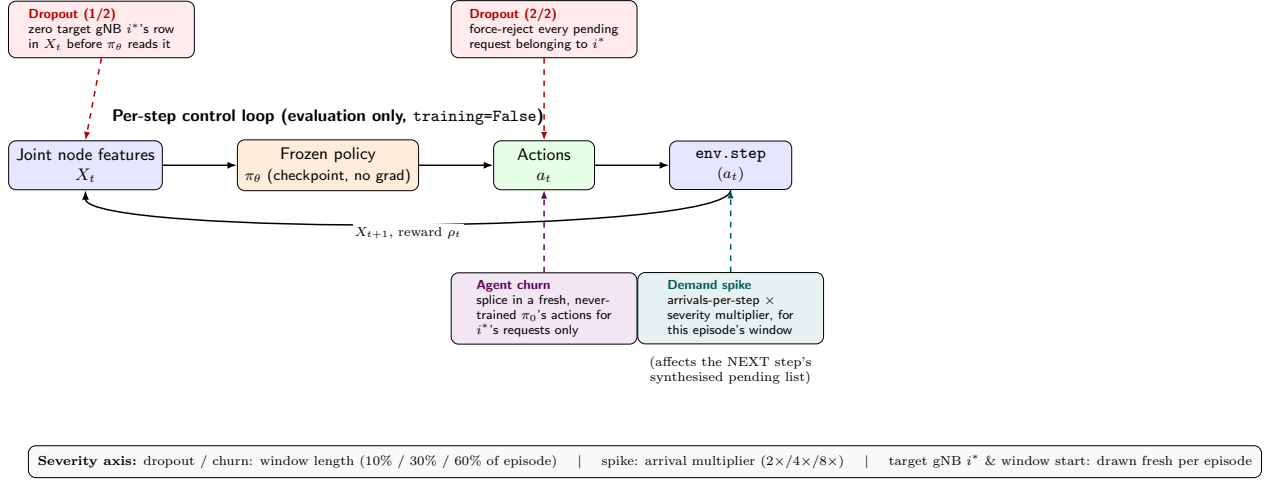


Fig. 6. Where each M4 disruption kind intercepts the frozen policy’s per-step evaluation loop. Dropout acts twice (corrupting the target gNB’s input row, then separately forcing that gNB’s own requests to reject, so a policy cannot simply learn around the corrupted feature); agent churn substitutes a fresh, never-trained policy’s actions for the target gNB only, leaving every other agent’s decisions untouched; demand spike is entirely external to the policy, acting on the arrival process the environment synthesises for the following step. All three are evaluation-only perturbations – the policy itself is never updated.

rather than retyped from either sample’s summary numbers. Panel (a)’s three architecture/federation comparisons and panel (b)’s dropout/churn costs for GAT-CTDE and the federated arm all keep the same sign and a broadly overlapping confidence interval across samples – visually, the committed and fresh estimates sit close together on every row except one. The exception is independent DQN’s churn cost (panel (b), bottom row, shaded): both samples agree the cost is positive (churn never helps), but the committed sample’s interval reaches down toward zero while the fresh sample’s sits clearly higher and to the right of every other churn estimate in the panel – the same reversal Section VIII-D reports in numbers, here shown as the one row that visibly does not track its own sample-to-sample agreement the other six rows show.

IX. DISCUSSION

Three findings in this paper are, we think, more broadly useful than the specific architecture: first, that a class-imbalanced, shared Q-head can silently suppress a minority slice’s learning signal in a multi-slice admission controller, and that per-slice heads are a simple, verifiable fix; second, that an SLA-compliance metric which scores decisions only by whether they eventually keep a slice within its margin – not by whether the decision itself was correct given the information available at the time – can actively reward a degenerate policy over a correct one whenever the correct action cannot recover an already-stressed slice’s margin. Any admission-control evaluation in a stress regime where this asymmetry holds should report a decision-level correctness metric alongside, not instead of, an outcome-level compliance metric.

Third, that reproducing a result on the same seed and replicating it on an independent one answer different questions, and only the second caught a real error in this paper’s own analysis. Every number in this paper was checked for exact, same-seed reproducibility (an eval-log append bug and an unseeded weight-initialisation path, Section IV-E, both surfaced and were fixed this way, neither by inspection alone), which

confirms the pipeline is deterministic but says nothing about whether a finding is real or a property of one particular sample of seeds. Only the independent-seed replication in Section VIII caught the latter: a claimed architectural immunity to agent churn that read as “mechanistically sensible” on one 10-seed sample was directly contradicted by a second, disjoint one. We would not have caught this by re-running the same seeds again, however many times, and we think any paper reporting an architecture-specific claim from a single seed sample – ours included, elsewhere, before this check – should weigh how much of its evidence is reproduction versus replication.

We are explicit about what remains offline-only. Every result in this paper uses the 3-gNB offline stress environment; none of it is a live claim, and Section II explains why we do not believe an offline-trained ranking here would transfer live without separate verification. Federated learning across three gNBs under one operator’s cluster is also, as configured, a synthetic privacy scenario – genuine cross-organisation data separation would need multiple operators unwilling to pool raw data, which nothing in our current setup represents. We lead with the operational coordination benefit of periodic, not per-step, synchronisation (a real constraint for disaggregated near-real-time RAN Intelligent Controllers (near-RT RICs) with limited backhaul, even under one operator) rather than with a privacy motivation we cannot substantiate at this scale.

X. CONCLUSION AND FUTURE WORK

We built and iteratively fixed a GAT-CTDE architecture for coordinated multi-gNB slice admission control, found and diagnosed three failures a standard compliance metric, an ungarded append-only log, and an unseeded weight-initialisation path could not have caught on their own – a training collapse, a hidden eval-data contamination bug, and a same-seed-only reproducibility gap – and showed that on a correctness-aware reward metric the architecture delivers a decisive, statistically significant improvement over an independent-learner ablation, though its edge over our existing

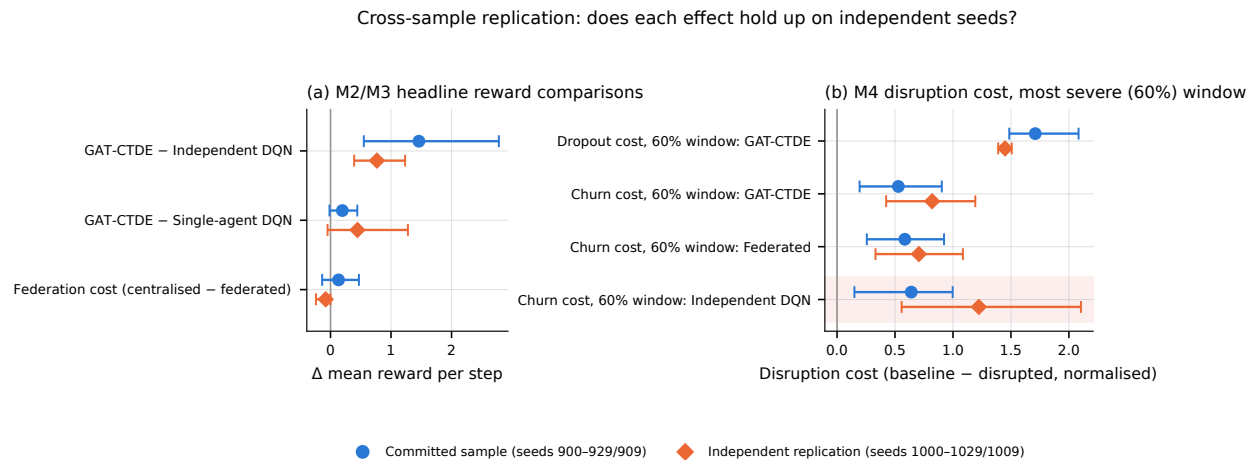


Fig. 7. Every headline paired comparison in this paper, committed sample (seeds 900–929/909) against the independent replication sample (seeds 1000–1029/1009), recomputed from raw logs for both. (a) M2/M3 reward comparisons. (b) M4 disruption cost at the most severe (60%) window. The bottom row of (b), independent DQN’s churn cost, is the one comparison whose two samples visibly disagree on more than point-estimate noise – the finding retracted in Section VIII-D.

single-agent baseline specifically is small and not decisive at this sample size. Extending the architecture to federated training with differential privacy, we found federation to add no measurable reward cost and privacy costly only past a real threshold in decision precision, not a smooth trade-off. Evaluating the same frozen policies under mid-episode gNB dropout, demand spikes, and agent churn, we confirmed the threshold prediction this paper’s own DP-noise finding suggested: dropout and churn both produce a real, accelerating cost with severity rather than graceful degradation, while demand spikes do not threshold at all on the metric that matters, a genuinely different failure mode. Cross-checking every result against an independent seed sample, not just the same seeds reproduced again, caught one architectural claim (that independent DQN is immune to agent churn) that a single sample had made look confirmed but was not – retracted before it could appear as a finding, with the correction reported alongside it rather than silently.

Everything in this paper is offline-only, and validating whether any of these architectural or privacy-cost rankings survive contact with a live testbed – as our single-agent work already did, and as this offline environment’s own known live-fidelity gap (Section II) says we should not assume – is the most direct remaining step. Two narrower questions this paper’s own results raise are also still open: whether the churn-cost gap between coordination-dependent and independent architectures is itself statistically real (no formal paired test between architectures’ disruption costs was run here, only each architecture against its own undisrupted baseline), and why the privacy threshold’s exact location shifted between our two seed samples (Section VIII) – whether that reflects genuine sensitivity to which ten seeds are drawn, or a factor this paper has not identified.

ACKNOWLEDGMENT

TODO(MANOJ): Same acknowledgment text as paper_conf/main.tex, or updated for this paper – please fill in.

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